

Precept 2 - Solutions

Exercise 1 (Monty Hall Problem)

This problem is about a fictional game show in which you, the player, pick a door and get whatever is behind it. One door has a shiny new sports car, the other two have goats. Suppose that you're on this show and that after you make your pick the host (Monty Hall) opens one of the other doors to reveal a goat. Before your chosen door is opened you have the option of switching to the other, yet unopened door. Do you take it?

Solution:

We start by clarifying one common source of confusion. That is, whether or not Monty knows what is behind the doors. We'll go with the interpretation that he does know and will never open the door showing the car. Many people argue that the two unopened doors are the same so they each will contain the car with probability $1/2$, and hence there is no point in switching. We show that this naive reasoning is incorrect.

Without loss of generality, let us assume that you pick door #1 and Monty opens door #2 to reveal a goat. We are interested in the probability that the car is behind door #1 upon the revelation that there is a goat behind door #2.

Let's define the following events:

A_1 := Car is behind door #1.

A_2 := Car is behind door #2.

A_3 := Car is behind door #3.

B := Monty shows us a goat behind door #2.

Our desired probability is therefore $\mathbb{P}(A_1|B)$. Before doing any computations, let us extract from the story above what we know.

Firstly, at the beginning of the game, before any decisions are made, the car is equally likely to be behind any one of the three doors. So we have that $\mathbb{P}(A_1) = \mathbb{P}(A_2) = \mathbb{P}(A_3) = 1/3$.

Since we selected door #1, Monty can only open door #2 or door #3 to reveal a goat.

If the car is indeed behind door #1, then Monty will open door #2 half the time. So $\mathbb{P}(B|A_1) = 1/2$.

If the car is indeed behind door #2, Monty will never open it. So $\mathbb{P}(B|A_2) = 0$.

Finally, if the car is indeed behind door #3 and we picked door #1, then Monty will always show you door #2. So $\mathbb{P}(B|A_3) = 1$.

Now that we are done extracting mathematical information from the story, we are ready to do some computation. We want to compute

$$\mathbb{P}(A_1|B) = \frac{\mathbb{P}(A_1 \cap B)}{\mathbb{P}(B)}.$$

From Bayes Theorem, we know that this can also be computed as

$$\mathbb{P}(A_1|B) = \frac{\mathbb{P}(B|A_1)\mathbb{P}(A_1)}{\mathbb{P}(B|A_1)\mathbb{P}(A_1) + \mathbb{P}(B|A_2)\mathbb{P}(A_2) + \mathbb{P}(B|A_3)\mathbb{P}(A_3)}.$$

Now we are in business because we actually know these probabilities. Hence

$$\mathbb{P}(A_1|B) = \frac{1/2 \cdot 1/3}{1/2 \cdot 1/3 + 0 \cdot 1/3 + 1 \cdot 1/3} = \frac{1/6}{1/2} = \frac{1}{3}.$$

Therefore probability that car is NOT behind door #1 given a goat is behind door #2 is $\mathbb{P}(A_1^c|B) = 1 - \mathbb{P}(A_1|B) = 2/3$.

Therefore it is worthwhile to switch since our probability of winning the car jumps from $1/3$ to $2/3$.

As mentioned earlier, there is a common misconception that it doesn't matter if you switch or not: you have a $1/2$ chance of winning no matter what you do. This is just as though Monty had started the contest with one door open. This ignores the fact that your choice of door influences Monty's choice. If you picked a door without a car behind it then he was forced to show you the only goat remaining. If you picked the car then he could choose either goat to show you.

Exercise 2 (Communication through a Noisy Channel)

A source transmits a message (a string of symbols) through a noisy communication channel. Each symbol is 0 or 1 with probability p and $1 - p$, respectively. And each symbol is received incorrectly with probability ϵ_0 and ϵ_1 , respectively (see Figure 1). Errors in different symbol transmissions are independent.

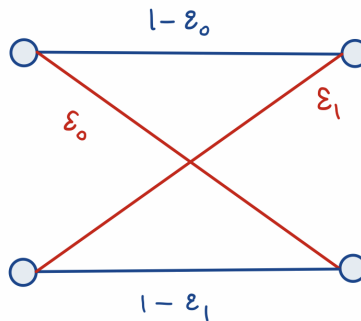


Figure 1: Probabilities in a binary communication channel

(a) What is the probability that the k -th symbol is received correctly?

- (b) What is the probability that the string of symbols 1011 is received correctly?
- (c) To improve reliability, each symbol is transmitted three times and the received string is decoded by majority rule. In other words, a 0 (or 1) is transmitted as 000 (or 111) and it is decoded at the receiver if and only if the received three-symbol string contains at least two 0s (or two 1s). What is the probability that a 0 is correctly decoded?
- (d) Suppose the three-symbol scheme is used. What is the probability that a symbol was 0 given that the received string is 101?

Solution:

- (a) Define the following events: S_0 denote the event that 0 was sent in the k -th symbol. Similarly for S_1 . R_0 denotes the event that 0 was received in the k -th symbol. Similarly for R_1 . Let A to be the event that the k -th symbol is received correctly. Then use law of total probability, we have

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A|S_0)\mathbb{P}(S_0) + \mathbb{P}(A|S_1)\mathbb{P}(S_1) \\ &= \mathbb{P}(R_0|S_0)\mathbb{P}(S_0) + \mathbb{P}(R_1|S_1)\mathbb{P}(S_1) \\ &= \boxed{(1 - \epsilon_0)p + (1 - \epsilon_1)(1 - p)}\end{aligned}$$

- (b) By the problem definition, we know that the probability that 1 is received correctly with probability $(1 - \epsilon_1)$ and 0 is received correctly with probability $(1 - \epsilon_0)$. Since errors in different transmissions are independent, we get

$$\begin{aligned}\mathbb{P}(\text{1011 is received correctly}) &= \mathbb{P}(\text{1 is received correctly})^3 \times \mathbb{P}(\text{0 is received correctly}) \\ &= \boxed{(1 - \epsilon_1)^3(1 - \epsilon_0)}\end{aligned}$$

- (c) In words, we know that 0 is correctly decoded if and only if 2 or more 0's are received. Hence define A to be the event that 0 is correctly decoded, B to be the event 2 zeros are received given 0 is sent, and C to be the event that 3 zeros are received given 0 is sent. Then we have

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(B) + \mathbb{P}(C) \\ &= \binom{3}{2}(1 - \epsilon_0)^2\epsilon_0 + \binom{3}{3}(1 - \epsilon_0)^3 \\ &= \boxed{3(1 - \epsilon_0)^2\epsilon_0 + (1 - \epsilon_0)^3}\end{aligned}$$

- (d) Let A be the event that 0 is sent and let B be the event that 101 is received. Then by Baye's rule, we have

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}$$

Term by term, we have $\mathbb{P}(A) = p$, $\mathbb{P}(B|A) = \epsilon_0^2(1 - \epsilon_0)$. By law of total probability we have

$$\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|A^c)\mathbb{P}(A^c) = p\epsilon_0^2(1 - \epsilon_0) + (1 - p)\epsilon_1(1 - \epsilon_1)^2.$$

In all, we have

$$\mathbb{P}(A|B) = \frac{p\epsilon_0^2(1 - \epsilon_0)}{p\epsilon_0^2(1 - \epsilon_0) + (1 - p)\epsilon_1(1 - \epsilon_1)^2}$$

Exercise 3 (Taxi Cab Paradox)

Overnight a Taxi in New York City is involved in a hit and run accident. There are two taxi companies in the city, one with yellow cars and one with green cars. The yellow cab company has 85% of taxis in the city and the green cab company has 15%. There was one witness to the hit and run, who thinks that the taxi was green. In order to determine the reliability of the witness the court ordered a test and found that the witness correctly identified the color of the taxi 80% of the time, and got it wrong 20% of the time. What is the probability that the taxi involved in the hit and run accident was green?

Solution:

We should first discuss what we are naturally inclined to believe. The witness is accurate with probability 0.8, so why not say that there was an 80% chance that the taxi was green? The answer is that *without* the witness we would have said it was only 15%, so this fact should act as an anchor to our estimate when we adjust our beliefs based on what the witness said. This process is known as *Bayesian Inference*.

What is our sample space? There are a total of four outcomes. One pair of outcomes for each color, and one pair determining whether the witness was right or wrong. Let our probability space be $\Omega = \{YR, YW, GR, GW\}$, where the first letter is the color and the second is right or wrong.

What is our probability? We assume that the witness is equally reliable for each type of car, so that the events $A = \{YR, GR\}$ and $B = \{YW, YR\}$ are independent. That is, the event "right" and the event "yellow" are independent. We know that $\mathbb{P}(A) = 0.8$ and $\mathbb{P}(B) = 0.85$.

How do we find the answer? Consider the event in which the witness claims the car involved in the accident was green. This event is $E = \{GR, YW\}$. The conditional probability that the car was green is

$$\mathbb{P}(B^C|E) = \mathbb{P}(B^C \cap E) / \mathbb{P}(E) = \mathbb{P}(\{GR\}) / \mathbb{P}(\{GR, YW\}).$$

We have $\mathbb{P}(\{GR\}) = \mathbb{P}(\{GR, GW\}) \mathbb{P}(\{GR, YR\}) = 0.15 \cdot 0.8 = 0.12$ by independence. We also have $\mathbb{P}(\{GR, YW\}) = \mathbb{P}(\{GR\}) + \mathbb{P}(\{YW\}) = 0.12 + 0.85 \cdot 0.2 = 0.29$. Then the probability that the car in the accident was green is $0.12/0.29 = 0.414$. So it is *more* likely than not that the witness is wrong and the car is yellow.